GMI German-Malaysian Institute

DIPLOMA PROGRAMME

LAB WORK 1

Academic Period: January 2026

Program	:	**CYBER SECURITY TECHNOLOGY (CBS)**
Course	:	**NETWORK MANAGEMENT & MONITORING SYSTEMS**
Code	:	**NTC 1052**
Total Marks	:	**20 MARKS**

Title	:	**INTRODUCTION TO NAGIOS CORE**

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Semester/Trade/Group	:	SEMESTER 5 / DCBS / DCBS 1
Date	:	7/7/2026

NO.	CRITERIA	EXCELLENT [4]	GOOD [3]	SATISFACTORY [2]	WEAK/ NOT MET [1]	MARKS
INDIVIDUAL COMPONENT (TOTAL MARK = 20)						
1	**Understanding of Lab Objectives**	Clearly explains the purpose, scope, and objectives of the lab work	Explains objectives with minor gaps	Basic understanding of objectives	Objectives misunderstood or not explained	
2	**Demonstration of Lab work Execution**	Demonstrates lab tasks correctly, confidently, and independently	Demonstration mostly correct with minor errors	Demonstration partially correct or requires guidance	Unable to demonstrate lab tasks correctly	
3	**Accuracy of Findings / Results**	Findings are accurate, relevant, and clearly linked to the task	Mostly accurate findings with minor errors	Limited or partially correct findings	Unable to demonstrate lab tasks correctly	
4	**Fluency and Clarity of Explanation**	Explanation is clear, fluent, well-structured, and uses correct technical terms	Generally clear with minor language or structure issues	Explanation is basic and sometimes unclear	Poor, unclear, or disorganized explanation	
5	**Question & Answer (Q&A)**	Answers all questions confidently and accurately	Answers most questions correctly	Answers some questions with hesitation some questions with hesitation	Unable to answer questions correctly	
				TOTAL	20	

1. What is Nagios Core:

1. Nagios® Core™ is an Open Source system and network monitoring application. It watches hosts and services that you specify, alerting you when things go bad and when they get better.

2. Originally runs under Linux OS (preferably SuSe Linux or Open SuSe).

3. Features in Nagios included:

a. Monitoring of network services (SMTP, POP3, HTTP, NNTP, PING, etc.)
b. Monitoring of host resources (processor load, disk usage, etc.)
c. Simple plugin design that allows users to easily develop their own service checks
d. Parallelized service checks
e. Ability to define network host hierarchy using "parent" hosts, allowing detection of and distinction between hosts that are down and those that are unreachable
f. Contact notifications when service or host problems occur and get resolved (via email, pager, or user-defined method)
g. Ability to define event handlers to be run during service or host events for proactive problem resolution
h. Automatic log file rotation
i. Support for implementing redundant monitoring hosts
j. Optional web interface for viewing current network status, notification and problem history, log file, etc.

4. Requirements:
   a. WebServer (apache)
   b. Gd library (1.6.3 or higher)

2. Go to demo version of nagios. http://nagioscore.demos.nagios.com/
   Choose 'Login To Nagios Core'

Advanced User Access: An advanced user account that can see and control all hosts and services.

You can access the demo with different accounts to get a different view of the monitoring system.

- **Administrator Access:** An account with administrative privileges. ○ Username: nagiosadmin ○ Password: nagiosadmin

Read-Only User Access: A user account that can view all hosts and services, but not make any changes or issue commands.

- Username: readonly
- Password: readonly

- Username: advanced
- Password: advanced

Normal User Access: A sample "normal" user account that has rights to view only a subset of all hosts and services.

- Username: jdoe
- Password: jdoe

3. Answer the question below. Stated any differences of view between each users below
    a. Username: nagiosadmin
    b. Username: readonly
    c. Username: advance
    d. Username: jdoe

How to install Nagios in openSUSE

To install Nagios server monitoring tool in OpenSUSE

In this article we will learn how to install Nagios in openSUSE. It is a free source application that monitors system, networks and infrastructure. It provides monitoring and alerting services for switches, applications, servers, and services.

To install pre-request packages

First install the pre-request packages by utilizing the following command.

```
opensuseleap15:/home/openSUSE/Downloads # zypper install apache2 php7
gcc glibc  gd wget perl make apache2-mod_php7

Retrieving repository ' openSUSE-13.2-Non-Oss' metadata
...................[done]

Building repository ' openSUSE-13.2-Non-Oss' cache
.........................[done]

Retrieving repository ' openSUSE-13.2-Oss' metadata
.........................[done]

.

.

.

(30/31) Installing: apache2-prefork-2.4.10-31.1
...........................[done]

(31/31) Installing: apache2-mod_php5-5.6.1-72.1
...........................[done]
```

To check and restart the deleted files, you can use ' zypper ps' to list these programs.
Then create user and group for nagios.

```
opensuseleap15:/home/openSUSE/Downloads # useradd nagios

opensuseleap15:/home/openSUSE/Downloads # groupadd ngadmin
```

Once you have created the user, add those user into the nagios group.

```
opensuseleap15:/home/openSUSE/Downloads # usermod -a -G ngadmin nagios

opensuseleap15:/home/openSUSE/Downloads # usermod -a -G ngadmin wwwrun
```

To download and install Nagios

Download the nagios package by running the following command.

```
opensuseleap15:/home/openSUSE/Downloads # wget -O nagioscore.tar.gz
https://github.com/NagiosEnterprises/nagioscore/archive/nagios4.4.5.tar
.gz
```

Once the download is completed, extract the downloaded tar package.

```
opensuseleap15:/home/openSUSE/Downloads # tar  xzf nagios-4.4.5.tar.gz

cd nagioscore-nagios-4.4.5
```

Now compile to install the Nagios.

```
                                                                    #
opensuseleap15:/home/openSUSE/Downloads/nagioscore-nagios-4.4.5
./configure --with-nagios-group=nagios --with-command-group=nagcmd

checking for a BSD-compatible install... /usr/bin/install -c
```

```
checking build system type... x86_64-unknown-linux-gnu

checking host system type... x86_64-unknown-linux-gnu

checking for gcc... gcc

.

.

.
```

```
Traceroute (used by WAP): /usr/sbin/traceroute Review the

options above for accuracy. If they look okay,

type ' make all'  to compile the main program and CGIs.
```

Run the following command to install the Nagios package.

```
opensuseleap15:/home/openSUSE/Downloads/nagioscore-nagios-4.4.5 # make
all

cd ./base & &  make

make[1]: Entering directory ' /home/user1/nagios-4.4.5/base'

gcc -Wall -I.. -g -O2 -DHAVE_CONFIG_H -DNSCORE -c -o nagios.o nagios.c

gcc -Wall -I.. -g -O2 -DHAVE_CONFIG_H -DNSCORE -c -o broker.o broker.c

.

.
```

```
. For more information on obtaining support for Nagios,

visit:



http://support.nagios.com

************************************************************

Enjoy.

opensuseleap15:/home/openSUSE/Downloads/nagioscore-nagios-4.4.5 # make

install

opensuseleap15:/home/openSUSE/Downloads/nagioscore-nagios-4.4.5 # make
install-init
```

```
/usr/bin/install -c -m 755 -d -o root -g root /etc/rc.d



/usr/bin/install -c -m 755 -o root -g root daemon-init /etc/rc.d/nagios



*** Init script installed ***
```

After installing Nagios, run the make install-config command.

```
opensuseleap15:/home/openSUSE/Downloads/nagioscore-nagios-4.4.5 # make

install-config
```

```
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/etc
```

```
/usr/bin/install -c -m 775 -o nagios -g nagios -d
/usr/local/nagios/etc/objects

/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-
config/nagios.cfg /usr/local/nagios/etc/nagios.cfg

/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/cgi.cfg
/usr/local/nagios/etc/cgi.cfg

.

.

.

/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-
config/template-object/switch.cfg

/usr/local/nagios/etc/objects/switch.cfg


*** Config files installed
```

Then run the make install-commandmode command.

```
opensuseleap15:/home/openSUSE/Downloads/nagioscore-nagios-4.4.5 # make
install-commandmode

/usr/bin/install -c -m 775 -o nagios -g nagcmd -d
/usr/local/nagios/var/rw
```

```
chmod g+s /usr/local/nagios/var/rw
```

```
*** External command directory configured ***
```

Finally use the below command for the webconf installation.

```
opensuseleap15:/home/openSUSE/Downloads/nagioscore-nagios-4.4.5 # make
install-webconf

/usr/bin/install       -c       -m       644       sample-config/httpd.conf
/etc/apache2/conf.d/nagios.conf
```

```
*** Nagios/Apache conf file
```

Verify the nagios access details as shown below.

```
(cd.. many time to move out from current directory)

opensuseleap15:~ # vi /usr/local/nagios/etc/objects/contacts.cfg
```

After verifying the Nagios access details, create password for nagios login and restart the apache services.

```
opensuseleap15:~  #  htpasswd2  -c  /usr/local/nagios/etc/htpasswd.users
nagiosadmin New password:1
```

```
Re-type new password:1
```

Adding password for user nagiosadmin

```
opensuseleap15:~ # systemctl restart apache2

opensuseleap15:~ # systemctl status apache2

opensuseleap15:~ # systemctl restart nagios

opensuseleap15:~ # systemctl status nagios



(status = to check current status of the service)
```

STOP HERE..

To be continue...

Installing The Nagios Plugins

sudo zypper --non-interactive install autoconf gcc glibc libgcrypt-devel make libopenssl-devel wget gettext gettext-runtime automake net-snmp perl-Net-SNMP

Utilize the following command for downloading the Nagios plugin.

```
opensuseleap15:/home/openSUSE/Downloads # wget --no-check-certificate O
      nagios-plugins.tar.gz
```

```
https://github.com/nagios-plugins/nagiosplugins/archive/release-2.2.1.tar.gz

tar zxf nagios-plugins.tar.gz

nagios-plugins-2.2.1.tar.gz
100%[=============================================> ] 2.54M 581KB/s
in 5.1s



2016-09-16 19:42:53 (512 KB/s) - ‘ nagios-plugins-2.2.1.tar.gz’  saved
[2659772/2659772]
```

Run the following command to extract the nagios plugin.

```
opensuseleap15:/home/openSUSE/Downloads # tar -xvf nagios-plugins-2.2.1.tar.gz

-2.
```

After extraction, compile and install the nagios plugin.

```
opensuseleap15:/home/openSUSE/Downloads/nagios-plugins-2.2.1#
./configure --with-nagios-user=nagios --with-nagios-group=nagios

checking for a BSD-compatible install... /usr/bin/install -c

checking whether build environment is sane... yes

checking for a thread-safe mkdir -p... /usr/bin/mkdir -p
```

```
checking for gawk... gawk

checking whether make sets $(MAKE)... yes

.
```

```
.

.

--with-perl: /usr/bin/perl

--enable-perl-modules: no

--with-cgiurl: /nagios/cgi-bin
```

```
--with-trusted-path: /bin:/sbin:/usr/bin:/usr/sbin

--enable-libtap: no
```

Use the following command to start installing the Nagios plugin.

```
opensuseleap15:/home/openSUSE/Downloads/nagios-plugins-2.2.1# make
```

After the plugins are installed, check the nagios configuration file.

```
opensuseleap15:/home/openSUSE/Downloads/nagios-plugins-2.2.1#

/usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg

Nagios Core 4.0.8

Copyright (c) 2009-present Nagios Core Development Team and Community
```

```
Contributors Copyright (c) 1999-2009 Ethan Galstad

Last Modified: 08-12-2014

.
```

```
.

.

Total Warnings: 0

Total Errors: 0
```

```
Things look okay - No serious problems were detected during the

preflight check
```

Then Start the nagios service

```
opensuseleap15:/home/openSUSE/Downloads/nagios-plugins-2.2.1# systemctl

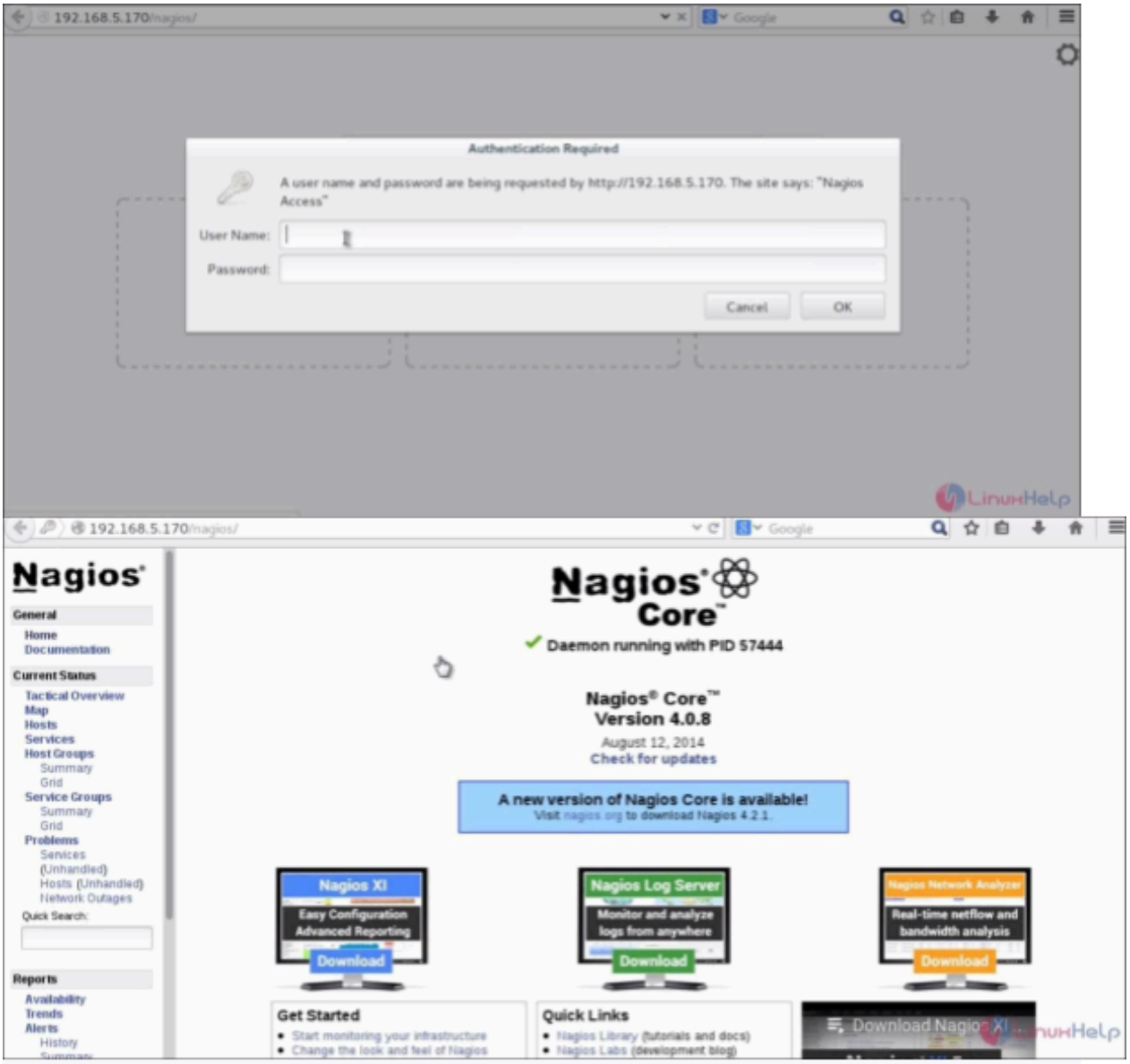
start nagios
```

You can also use the following command to start the nagios on boot itself.

```
opensuseleap15:/home/openSUSE/Downloads/nagios-plugins-2.2.1# chkconfig

nagios on
```

Trigger the browser with the following URL http://< IP_address> /nagios and enter the user credentials to login into the Nagios.



Now you can start checking the status of your server.


192.168.5.170/nagios/
Google
Nagios
General
Home
Documentation
Current Status
Tactical Overview
Map
Hosts
Services
Host Groups
Summary
Grid
Service Groups
Summary
Grid
Problems
Services
(Unhandled)
Hosts (Unhandled)
Network Outages
Quick Search:
Reports
Availability
Trends
Alerts
History
Summary
Current Network Status
Last Updated: Fri Sep 16 19:59:37 IST 2016
Updated every 90 seconds
Nagios® Core™ 4.0.8 - www.nagios.org
Logged in as nagiosadmin
View History For all hosts
View Notifications For All Hosts
View Host Status Detail For All Hosts
Host Status Totals
Up Down Unreachable Pending
1 0 0 0
All Problems All Types
0 1
Service Status Totals
Ok Warning Unknown Critical Pending
6 1 0 1 0
All Problems All Types
2 8
Service Status Details For All Hosts
Limit Results: 100
Host Service Status Last Check Duration Attempt Status Information
localhost Current Load OK 09-16-2016 19:57:41 0d 0h 6m 56s 1/4 OK - load average: 0.16, 0.22, 0.21
Current Users OK 09-16-2016 19:58:19 0d 0h 6m 18s 1/4 USERS OK - 2 users currently logged in
HTTP WARNING 09-16-2016 19:56:56 0d 0h 5m 41s 4/4 HTTP WARNING: HTTP/1.1 403 Forbidden - 1249 bytes in 0.002 second response time
PING OK 09-16-2016 19:54:34 0d 0h 5m 3s 1/4 PING OK - Packet loss = 0%, RTA = 0.05 ms
Root Partition OK 09-16-2016 19:55:11 0d 0h 4m 26s 1/4 DISK OK - free space: / 9133 MB (68% inode=82%):
SSH CRITICAL 09-16-2016 19:58:49 0d 0h 3m 48s 4/4 connect to address 127.0.0.1 and port 22: Connection refused
Swap Usage OK 09-16-2016 19:56:26 0d 0h 3m 11s 1/4 SWAP OK - 99% free (3000 MB out of 3059 MB)
Total Processes OK 09-16-2016 19:57:04 0d 0h 2m 33s 1/4 PROCS OK: 248 processes with STATE = RSZDT
Results 1 - 8 of 8 Matching Services
LinuxHelp

Proof of Concept :

Nagios was set up using openSUSE Leap 16.0 on VMware Workstation Pro and can be accessed either with localhost or using host computer.

